Valuation separation and signed certificates

Explicit Diophantine forms of Con(ZFC)

AI-assisted research prepared for Vladimir Reshetnikov
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Abstract

We give two explicit improvements to Diophantine representations of syntactic ZFC refutability. The first is an integer-coefficient polynomial in twelve unrestricted integer variables, of exact degree 2400, evaluated by 215 live binary arithmetic operations. It improves the inspected twelve-integer construction of degree 4832 and 248 operations. A companion has degree 2912 and 209 operations. The main observation is that the existing interpolated Pell certificate already separates its radicals at the prime 3. Constant weights replace growing polynomial weights. Retaining the auxiliary Pell root as a rational square root, then clearing its denominator inside a norm, gives a second degree reduction. A classical intersective quintic can replace the previous monotone carrier.

The second construction is a fully expanded quartic in 1,190 unrestricted integers, obtained from the recent synchronized direct proof-certificate construction. Its coefficients are ordinary decimal integers of absolute value at most 83,900,628. It uses the calibrated numerical proof calculus directly, independently of K215 and Jones's universal polynomial.

For each supplied equation, composing the proved arithmetic transformations with the identified inherited ordinary calibration proofs establishes, within ZFC, that absence of solutions is equivalent to Con(ZFC). Neither consistency nor inconsistency is assumed. These are ordinary mathematical proofs and reproducible finite certificates; no new proof-assistant derivation of the complete internal ZFC biconditional is claimed. The contribution includes exact equations, coefficient recipes, independent checks, source provenance, and further research questions.

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1 Results, comparison, and meaning of the claim

Fix the usual syntactic statement

$$\mathrm{Con}(\mathrm{ZFC}) \equiv \neg\exists p \in \mathbb{N}\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot).$$

The task is to specify a fixed polynomial P with integer coefficients and prove

$$\mathrm{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathbb{Z}^n\ P(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

Proving this equivalence does not prove either side. In particular, the construction does not evade the second incompleteness theorem by purporting to prove its own polynomial has no zero.

Theorem 1.1 (Supplied equations)**.** *The package specifies polynomials* $P_{2400}, P_{2912} \in \mathbb{Z}[j, C, z, h, r, v, w, T, c, n, t, s]$ *with exact degrees* 2400 *and* 2912*, respectively. Their delivered straight-line circuits use* 215 *and* 209 *live binary operations. Each satisfies* (1) *with* $n = 12$*, through the ordinary inherited native/compiler calibration identified in Section 9.*

It also supplies a literal quartic Q_{direct} *in 1,190 unrestricted integer variables with 26,411 collected nonzero monomials and coefficient height 83,900,628. This polynomial satisfies* (1) *through the independent numerical-calculus calibration in Section 8.*

Construction	Domain	Variables	Degree	Operations
Prior interpolated norm	$\mathbb{Z}$	12	4832	248
Prior shorter evaluator	$\mathbb{Z}$	12	16000	223
New cleared norm	$\mathbb{Z}$	12	**2400**	**215**
New integral-radicand norm	$\mathbb{Z}$	12	**2912**	**209**
Supplemental identical evaluator	$\mathbb{Z}_{>0}$	12	70	**98**
Prior identical polynomial	$\mathbb{Z}_{>0}$	12	70	100

Table 1: Counts for complete delivered circuits. A fixed scalar multiplication and a square each cost one multiplication. Fixed numeral leaves and reuse of an already computed value cost no operations. Domains are kept separate.

The reduction from 4832 to 2400 is 50.33% in total degree; the operation reduction from 248 to 215 is 13.31%. The companion uses six fewer operations at higher degree. Both new twelve-integer points improve the two inspected twelve-integer baselines in both measures.

There is no established scalar meaning of simplicity. The repository already contains ten-integer equations, nine-positive equations, and integer quartics with far fewer variables than the direct construction but very large fixed coefficients. We do not claim a global minimum or a new record for number of integer variables. The defensible claims are the particular equations, their proved transformations, and the measured improvements against identified parents.

The comparison includes recent incoming archives, not only the older collection README. The synchronized direct archive already contains a 522-natural-variable quartic and a 2088-integer companion; another incoming direct construction contains 1690 integer variables. The direct comparison includes both benchmarks [8, 9]. An earlier experiment improving an 8619-integer checkpoint was not selected as the principal result once these newer inputs were inspected.

2 A literal twelve-integer equation

2.1 Coordinates and fixed coefficients

All twelve independent variables range over all of $\mathbb{Z}$:

$$(j, C, z, h, r, v, w, T, c, n, t, s).$$

Uppercase C and lowercase c are distinct. Every other symbol below is a polynomial abbreviation, not an additional existential variable. Use the complete coherent source-34 six-tuple

$$(\beta, K_0, M_C, M_F, A_{\mathrm{cone}}, B_{\mathrm{cone}}), \qquad \beta = 2^{1064921} - 1, \quad A_{\mathrm{cone}} = 2^{1113}, \quad B_{\mathrm{cone}} = 2^{666104}.$$

Appendix A gives the bounded arithmetic recipe for all six numerals. The archive includes the entire finite tile table, machine table, recipe, and actual integer payloads. M_F already includes its prescribed shift. None of these numerals is a free program index or a value chosen by searching for a proof.

2.2 The arithmetic kernel

Define the following polynomial registers:

$$\begin{aligned} J &= 6j + 1, & q &= \beta J + 1, & W_0 &= 12w + 4, \\ X &= qW_0, & Y &= (2v + 1)q^3, & E &= XY, \\ a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H = (a + 2)^2 - 1. \end{aligned} \tag{2}$$

The transport and rank registers are

$$\begin{aligned} F &= (K_0 + W_0)C - z(q - 1), \\ R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\ k &= R + 1 + hE, & \eta &= c - Yk, & \zeta &= k - \eta. \end{aligned} \tag{3}$$

Set

$$\begin{aligned} G &= (A_{\mathrm{cone}}F)(B_{\mathrm{cone}}C) - (q - A_{\mathrm{cone}}F - B_{\mathrm{cone}}C)^2, \\ P &= G\eta\zeta - (t^2 - G - \eta\zeta + 1)^2, \\ D_{\mathrm{g}} &= (3P - 4)s^2 + 1, & I &= src^2, \\ D_{\mathrm{f}} &= 1 + \Delta I^2, & Q &= \Delta^2 I^2, & \delta &= Q - 1, \\ V &= D_{\mathrm{f}}(c - R + 4IT) - c, & C_{\mathrm{y}} &= QV^2 - 1, \\ D_\tau &= 1 + XY^2(XY^2 + 1)k^2, & D_{\mathrm{m}} &= 1 + \Delta c^2. \end{aligned} \tag{4}$$

The symbol P in (4) denotes the joint guard, not the final polynomial P_{2400}.

2.3 The smaller intersective carrier

Use the classical quintic

$$\pi(x) = (x^2 + 3)(x^3 - 19) = x^5 + 3x^3 - 19x^2 - 57. \tag{5}$$

Its intersectivity is a special case of Hyde, Lee, and Spearman [1]. The contribution here is its use in the new valuation-separated certificate; the quintic itself is not new. Write

$$b_0 = X + ac, \qquad u = HT - b_0, \qquad \Phi_H(x) = x^5 + 3H^2x^3 - 19H^3x^2 - 57H^5.$$

Thus $\Phi_H(x) = H^5\pi(x/H)$ as a polynomial identity, with no division required to evaluate it. In the formal quadratic algebra $d^2 = D_{\mathrm{m}}$, the remainder $\Phi_H(u + d) = A + Bd$ is computed by

$$\begin{aligned} Q_0 &= u^2 + D_{\mathrm{m}} + 3H^2, & C_0 &= u(u^2 + 3D_{\mathrm{m}}) - 19H^3, & C_1 &= 3u^2 + D_{\mathrm{m}}, \\ A &= Q_0C_0 + 2uD_{\mathrm{m}}C_1, & B &= Q_0C_1 + 2uC_0, \\ S &= IH^5, & Z &= Sn - A. \end{aligned} \tag{6}$$

The coefficient B is unrelated to any native radix.

2.4 The ordinary norm polynomial

Let $\mathcal{F}_{32}$ be the universal integer polynomial characterized by

$$\mathcal{F}_{32}(Z; A_0, \ldots, A_4) = \prod_{\epsilon \in \{-1,1\}^5} \left(Z - \sum_{i=0}^{4} \epsilon_i \sqrt{A_i} \right). \tag{7}$$

Changing the sign of any formal radical permutes the factors, so every surviving radical exponent is even. This defines an ordinary polynomial, not a square-root operation allowed in the delivered circuit. The primary equation is

$$\boxed{P_{2400} = \delta^{16} \mathcal{F}_{32} \left(Z;\, B^2 D_{\mathrm{m}},\, 9S^2 D_{\mathrm{f}},\, 81S^2 D_{\tau},\, 729S^2 D_{\mathrm{g}},\, \frac{S^2 C_{\mathrm{y}}}{\delta} \right) = 0.} \tag{8}$$

The degree of $\mathcal{F}_{32}$ in each radicand is at most 16, so the denominator clears completely over $\mathbb{Z}$. Section 6 gives an explicitly division-free interpretation. We also prove $\delta \neq 0$ on every integer assignment, so its rational interpretation creates no exceptional solutions. The lower-operation companion is

$$\boxed{P_{2912} = \mathcal{F}_{32} \left(Z;\, B^2 D_{\mathrm{m}},\, 9S^2 D_{\mathrm{f}},\, 81S^2 D_{\tau},\, 729S^2 D_{\mathrm{g}},\, S^2 \delta C_{\mathrm{y}} \right) = 0.} \tag{9}$$

Both equations use exactly the same twelve coordinates and fixed tuple.

3 Why the separating weights can be constant

Earlier constructions separated radicals by making each successive coefficient larger than all preceding terms. Such inequalities require polynomial weights, whose degrees accumulate in the final norm. Here the existing congruences provide separation at one fixed prime.

Lemma 3.1 (Unique least valuation)**.** *Let $r_1, \ldots, r_m$ be nonzero algebraic numbers with $r_i^2 \in \mathbb{Q}$. Suppose that in every subset containing at least two indices, the least value of $v_p(r_i^2)$ occurs at exactly one index and is even. Then no nonempty signed subset sum is zero. Consequently, if a signed sum containing every r_i is rational, all the r_i are rational.*

Proof. Fix a subset with at least two terms. Divide its roots by a rational power of p so that the uniquely least squared valuation becomes zero. The distinguished square is a unit of $\mathbb{Z}_{(p)}$; the other squares belong to $p\mathbb{Z}_{(p)}$. Form the product of the distinguished root minus every signed sum of the other roots. Sign invariance makes this product an integer polynomial in the squares. Modulo p, it is a power of the distinguished unit square, hence nonzero. None of its factors vanishes. A nonzero singleton cannot vanish either.

Now work in the multiquadratic splitting field generated by the roots. Apply any $\mathbb{Q}$-automorphism to the rational sum and subtract. The roots whose signs change give a vanishing signed subset sum after division by 2. The first part forces every root to be fixed by every automorphism, and hence to belong to the fixed field $\mathbb{Q}$. □

The lemma allows negative rational radicands. It does not assume the positivity guard before extracting it. Equivalently, one may group roots by rational square class and use the fact that rational numbers with a unique least p-adic valuation cannot sum to zero.

Lemma 3.2 (Intersectivity and the zero scale)**.** *The polynomial π in (5) has a root modulo every positive integer and no root in a real extension of $\mathbb{Q}$ of degree at most two.*

Proof. The factor x^2+3 has no real root, and x^3-19 is irreducible by Eisenstein at 19. A degree-three element cannot belong to an extension of degree one or two.

If $p \equiv 1 \pmod 3$, a primitive cube root of unity ω in $\mathbf{F}_p$ gives $(2\omega+1)^2 = -3$, a simple quadratic root. This includes $p = 19$. If $p \equiv 2 \pmod 3$, cubing permutes $\mathbf{F}_p^\times$. Since $p \nmid 19$, the cubic has a nonzero simple root. This also handles $p = 2$. Simple-root lifting supplies roots at every power of these primes. For $p = 3$, write $x = 1 + 3t$. The equation $x^3 = 19$ becomes

$$t + 3t^2 + 3t^3 = 2.$$

The left side minus two has root $t = 2$ and derivative 1 modulo 3. Its successive lifts give roots modulo every power of 3. The Chinese remainder theorem finishes. This is the $n = 19$ case of the classical classification [1]. □

Lemma 3.3 (Unconditional residues)**.** *For every integer assignment,*

$$q \equiv -1, \quad W_0 \equiv 1, \quad X \equiv -1, \quad a \equiv H \equiv \Delta \equiv 0 \pmod 3,$$
$$D_{\mathrm{m}} \equiv D_{\mathrm{f}} \equiv D_\tau \equiv 1, \quad Q \equiv 0, \quad \delta \equiv C_{\mathrm{y}} \equiv -1, \quad u \equiv B \equiv 1 \pmod 3.$$

Also $H \neq 0$, $\Delta > 1$, $D_{\mathrm{m}} \geq 1$, and $\delta \neq 0$.

Proof. The exponent 1064921 is odd, so $\beta \equiv 1 \pmod 3$. The residues follow directly. For the odd remainder,

$$B \equiv 5u^4 + 10u^2 D_{\mathrm{m}} + D_{\mathrm{m}}^2 \equiv 16 \equiv 1 \pmod 3.$$

Thus $B \neq 0$ without a monotonicity argument. The equation $4a + 3 = 0$ has no integer solution. Since J is odd and nonzero, $|q| \geq \beta - 1 \geq 16$, $|W_0| \geq 4$, $|X| \geq 64$, and $|Y| \geq 4096$. Hence $|a| \geq 63 \cdot 4096$, giving $\Delta > 1$. Finally $Q = (\Delta I)^2$ is either zero or greater than one. □

4 Soundness over unrestricted integers

The proof first excludes a zero multiplier. It then applies the valuation lemma before making any positivity assumption about the guard. This order is needed because all twelve coordinates are unrestricted integers.

4.1 The zero-multiplier boundary

Lemma 4.1. *Neither (8) nor (9) has an integer zero with $S = 0$. At every zero, I, s, r, c are nonzero and $Q > 1$.*

Proof. By Lemma 3.3, $H \neq 0$. Thus $S = 0$ implies $I = 0$, $Q = 0$, and $\delta = -1$. All scaled radicands except $B^2 D_{\mathrm{m}}$ vanish, and both final polynomials specialize to

$$\left(A^2 - B^2 D_{\mathrm{m}}\right)^{16}.$$

The two factors of the inner expression over $\mathbb{Q}(\sqrt{D_{\mathrm{m}}})$ are

$$A \pm B\sqrt{D_{\mathrm{m}}} = H^5 \pi\left(\frac{u \pm \sqrt{D_{\mathrm{m}}}}{H}\right).$$

Since $D_{\mathrm{m}} \geq 1$, both arguments belong to a real extension of $\mathbb{Q}$ of degree at most two. Lemma 3.2 says neither factor can vanish. Therefore $S \neq 0$. Now $I = src^2$ is nonzero, so each of s, r, c is nonzero; $Q = (\Delta I)^2 > 1$ follows from $\Delta > 1$. □

4.2 Root extraction without a sign assumption

Suppose first that $P_{2400} = 0$. The unconditional fact $\delta \neq 0$ allows the rational interpretation in (8). In a splitting field, one factor of the norm vanishes. After division by $S \neq 0$, the five radical terms, up to independent signs, are

$$\frac{B\sqrt{D_{\mathrm{m}}}}{S}, \qquad \sqrt{C_{\mathrm{y}}/\delta}, \qquad 3\sqrt{D_{\mathrm{f}}}, \qquad 9\sqrt{D_{\tau}}, \qquad 27\sqrt{D_{\mathrm{g}}}. \tag{10}$$

Their signed sum is the rational number $n - A/S$. The first four terms are nonzero by the residues in Lemma 3.3. Their squared 3-adic orders are, respectively,

$$-2v_3(S), \qquad 0, \qquad 2, \qquad 4. \tag{11}$$

Here $v_3(S) = v_3(I) + 5v_3(H) \geq 5$. If $D_{\mathrm{g}} \neq 0$, the last squared order is

$$6 + v_3(D_{\mathrm{g}}) \geq 6;$$

it need not be even. If $D_{\mathrm{g}} = 0$, omit the zero term. Every subset with at least two surviving terms has a unique even minimum, because any such subset contains one of the first four terms. Thus Lemma 3.1 makes all terms in (10) rational. This argument also covers $D_{\mathrm{g}} < 0$; no realness premise was used.

Since $B/S \neq 0$, it follows that $D_{\mathrm{m}}, D_{\mathrm{f}}, D_{\tau}, D_{\mathrm{g}}$ have rational square roots. A rational square root of an integer is an integer, so there are integers d, f, τ, g with

$$d^2 = D_{\mathrm{m}}, \qquad f^2 = D_{\mathrm{f}}, \qquad \tau^2 = D_{\tau}, \qquad g^2 = D_{\mathrm{g}}. \tag{12}$$

The auxiliary root is initially rational; Section 4.4 will prove it integral.

For $P_{2912} = 0$, replace the second term of (10) by $\sqrt{\delta C_{\mathrm{y}}}$. It again has squared 3-adic order zero, so the same proof supplies an integer $\widetilde{y}$ with $\widetilde{y}^2 = \delta C_{\mathrm{y}}$ and all four roots in (12).

4.3 The strict guards

The extracted square $g^2 = D_{\mathrm{g}}$ proves $D_{\mathrm{g}} \geq 0$. Since s is a nonzero integer, $P \leq 0$ would give

$$D_{\mathrm{g}} = (3P - 4)s^2 + 1 \leq -3.$$

Hence $P > 0$. We record the elementary guard principle to make this sign inference explicit.

Lemma 4.2 (Joint positivity)**.** *For integers A_1, A_2, there exists an integer t such that*

$$A_1A_2 - (t^2 - A_1 - A_2 + 1)^2 > 0$$

if and only if $A_1 > 0$ and $A_2 > 0$.

Proof. The inequality implies $A_1A_2 > 0$. If both factors were negative, then $t^2 - A_1 - A_2 + 1 > |A_1| + |A_2| \geq 2\sqrt{A_1A_2}$, a contradiction. Conversely, put $M = A_1 + A_2 - 1 \geq 1$. An integer square nearest to the integer M differs from M by an integer of absolute value strictly less than $\sqrt{M}$: between consecutive squares $a^2, (a+1)^2$, the nearest-square error is at most a, and at an endpoint it is zero. Since $M \leq A_1A_2$, this choice proves the strict inequality. □

Applying the forward implication to $A_1 = G$, $A_2 = \eta\zeta$ yields

$$G > 0, \qquad \eta\zeta > 0. \tag{13}$$

The determinant guard also restores the signs required by the inherited signed kernel. Set $U = A_{\mathrm{cone}}F$, $W = B_{\mathrm{cone}}C$, and $\alpha = q - F - C$. From $UW > (q - U - W)^2$ one obtains $UW > 0$.

If q had the opposite sign to U, W, the square would be at least $(|U| + |W|)^2 \geq 4UW$, impossible. Moreover

$$|q| > |U| + |W| - \sqrt{UW} \geq \tfrac{1}{2}(|U| + |W|).$$

Since both cone scales are at least two, F, C, α have the same nonzero sign as q.

For clarity, $R > 0$ already follows at this point. The authentic fixed tuple has $0 < M_C < \beta < M_F$. If $q > 0$, then $J > 0$ and

$$q(q - F) - C = q\alpha + (q - 1)C > 0.$$

Both summands in the definition of R are positive. If $q = -q_0 < 0$, write $C = -C_0$, $\alpha = -\alpha_0$, with $C_0, \alpha_0 > 0$. Then

$$q(q - F) - C = q_0(C_0 + \alpha_0) + C_0 > 0, \quad J < 0, \quad M_C - q_0 M_F < 0.$$

Again both summands defining R are positive. Thus no later rank argument is being used to justify this preliminary sign fact.

4.4 The rational auxiliary root is integral

Lemma 4.3 (Adjacent-coefficient descent)**.** *Let $Q > 1$ be an integer and let $V \in \mathbb{Z}$, $y \in \mathbb{Q}$ satisfy*

$$QV^2 - (Q - 1)y^2 = 1.$$

Then $y \in \mathbb{Z}$.

Proof. Changing signs makes V, y positive. The norm excludes $V = 0$. Write $y = u/b$ in lowest terms with $b > 0$. The equation implies $b^2 \mid Q - 1$, so $(Q - 1)y$ is integral. If $V > 1$, use

$$V' = (2Q - 1)V - 2(Q - 1)y,$$
$$y' = (2Q - 1)y - 2QV.$$

This transform preserves the norm, and V' is integral. Because $2Q - 1 \equiv 1 \pmod{b}$, the reduced denominator of y' is still b. The norm gives $y > V$ and $(Q - 1)y < QV$, hence $-V < V' < V$. The norm also excludes $V' = 0$. Taking absolute values therefore gives descent in the positive integer V without changing b. At $V = 1$ the equation gives $|y| = 1$, so $b = 1$. □

For the primary equation, the rational root y extracted from C_y/δ satisfies the lemma's equation because $\delta = Q - 1$ and $C_y = QV^2 - 1$. For the companion, take $y = \widetilde{y}/\delta$; its square is again C_y/δ. Lemma 4.1 supplies $Q > 1$ in both cases. Thus the same integer auxiliary Pell witness is recovered from both polynomials.

4.5 Recovering the main congruence

Absorb the signs of a vanishing conjugate factor into the integral roots. The primary equation gives

$$Sn = A + Bd + S(y + 3f + 9\tau + 27g). \tag{14}$$

For the companion, replace y on the right by $\widetilde{y}$. Consequently $H^5 \mid A + Bd$ in both cases. Since

$$A + Bd = \Phi_H(HT + d - b_0),$$

this restores exactly the missing congruence. Indeed, if $p^e \| H$ and $v_p(HT + d - b_0) = b < e$, the leading fifth power in

$$\Phi_H(z) = z^5 + 3H^2z^3 - 19H^3z^2 - 57H^5$$

has valuation $5b$, strictly below all three other terms. Divisibility by p^{5e} is impossible. Doing this for each prime divisor of H proves $H \mid HT + d - b_0$, and therefore

$$\gamma = \frac{d - b_0}{H} \in \mathbb{Z}, \qquad d = X + ac + \gamma H. \tag{15}$$

This introduces no independent coordinate; γ is reconstructed from an existing zero.

4.6 The precise inherited signed-kernel interface

The recovered equations are

$$\begin{aligned} d^2 - \Delta c^2 &= 1, & d &= X + ac + \gamma H, \\ f^2 - \Delta I^2 &= 1, & I &= src^2 \neq 0, \\ (\Delta I)^2 V^2 - ((\Delta I)^2 - 1)y^2 &= 1, & V &= f^2(c - R + 4IT) - c, \\ \tau^2 - XY^2(XY^2+1)k^2 &= 1, & G &> 0, \quad \eta\zeta > 0. \end{aligned} \tag{16}$$

The generalized signed-kernel proof used by the incoming twelve-variable construction is preserved as `integer12/proofs/INTERPOLATION12.md`, Sections 5–7, and its independent arithmetic audit as `integer12/proofs/INDEPENDENT_TWELVE_AUDIT.md`, Sections 2–4. They specify which ordinary rank, orientation, and native-decoding theorems are inherited. We give the normalized strong-rank argument here because it prevents an incorrect parity shortcut.

For an integer $A_* > 1$, define the Pell polynomials by

$$\chi_{A_*}(m) + \psi_{A_*}(m)\sqrt{A_*^2 - 1} = (A_* + \sqrt{A_*^2 - 1})^m, \qquad m \geq 0. \tag{17}$$

The inherited first/main rank bootstrap, applied to the first and last Pell equations and the strict guards in (16), gives

$$c_* = |c| = \psi_{A_*}(p), \qquad A_* = |a+2|, \qquad p \geq 25.$$

The normalized strong equation in (16) gives

$$|I| = \psi_{A_*}(m), \qquad |f| = \chi_{A_*}(m), \qquad m \geq 1.$$

This uses the usual integral Pell classification for $A_*^2 - 1$: $A_* + \sqrt{A_*^2 - 1}$ is the fundamental positive unit with integer coefficients, as follows by descent on the first coordinate.

Since $c_*^2 \mid |I|$, strong divisibility of the Lucas sequence ψ_{A_*} gives $p \mid m$. Write $m = p\ell$. Expanding the ℓth power of the unit at index p gives

$$\frac{\psi_{A_*}(p\ell)}{c_*} \equiv \ell \chi_{A_*}(p)^{\ell - 1} \pmod{c_*}.$$

The left side is divisible by c_*, and the Pell identity gives $\gcd(\chi_{A_*}(p), c_*) = 1$. Hence $c_* \mid \ell$ and, in particular, $pc_* \mid m$. These are the strong-rank inputs needed by the inherited signed auxiliary descent. They do not require I/c^2 to be odd.

For the auxiliary interface put $\rho = \Delta I$. The discarded exposed quotient coordinates are restored by the integer polynomials

$$\begin{aligned} o &= f(c - R + 4IT), \\ j_{\mathrm{aux}} &= \Delta s^2 r^2 c^3 (c - R) + 4 src D_{\mathrm{f}} T. \end{aligned}$$

Direct expansion gives

$$V = fo - c = cj_{\mathrm{aux}} - R, \qquad \rho^2 = \Delta(f^2 - 1).$$

Also $\rho^2 \equiv 1 - A_*^2 \pmod{|f|}$ and $\rho^2 \equiv 0 \pmod{c_*}$. In the auxiliary Pell classification, the first coordinate is $|\rho V|$. Its index is odd because

$$\chi_{|\rho|}(2j) \equiv (-1)^j \pmod{|\rho|}, \qquad |\rho| > 1,$$

whereas $\rho \mid \rho V$. This step likewise needs no oddness of ρ.

The inherited signed auxiliary step-down now yields $p = R$ and $2n_{\text{first}} = R + 1$; its main-root residue gives $X = \operatorname{sgn}(c)2^R$. In particular R is odd, so the fixed residue $X \equiv 2 \pmod 3$ selects $c > 0$ and $X = 2^R$. The inherited first-index sign identity gives $k > 0$, and $c/k = Y + \eta/k$, with $0 < \eta/k < 1$ and $|Y| > 1$, then gives $Y > 0$. Finally $q \mid 2^R$, $q \equiv 1 \pmod \beta$, and the inherited Mersenne-radix sign lemma give $q > 0$. The post-power native decoder and its fixed blank-input calibration supply a finite syntactic ZFC refutation. The relevant native proofs are retained under `integer12/inherited/root_fusion/provenance/`, including `PROOF44.md` and the `upstream_ordinary/` calibration documents. This is a composition of their ordinary theorems, rather than a claim that finite circuit checks establish the unbounded compiler.

Remark 4.4. An attempted restoration $r_{\text{old}} = (\Delta sr - 1)/2$ would be invalid when Δsr is even. The normalized rank argument above is the reason arbitrary nonzero multipliers are admissible. The proof does not silently import the older odd-multiplier coordinate chart.

5 Completeness and the interpolated residue

The only new completeness choice is a residue for the shorter quintic. We reproduce the interpolation mechanism because its quantifier order is essential: one strong Pell witness must be fixed before the requested residue class is selected.

The inherited canonical native construction supplies the outer data and main Pell roots with

$$\begin{gathered} A_* = a + 2 > 1 \text{ even}, \qquad R \equiv 3 \pmod 4, \\ c = \psi_{A_*}(R) > 0, \qquad d = \chi_{A_*}(R), \qquad \gamma = (d - X - ac)/H \in \mathbb{Z}, \\ G > 0, \qquad \eta\zeta > 0, \qquad \tau^2 = D_\tau. \end{gathered} \tag{18}$$

In particular $c \equiv R \pmod 4$, since the recurrence gives $\psi_{A_*}(j+2) \equiv -\psi_{A_*}(j) \pmod 4$ when A_* is even. The source-34 tuple's bounded recipe and the exact canonical hypotheses are recorded in Appendix A and the inherited coefficient proof. This is the canonical family used in the incoming interpolation theorem, not a restriction imposed on arbitrary zeros.

Lemma 5.1 (Full residue interpolation)**.** *Let $A_* > 1$ be even, $\Delta = A_*^2 - 1$, $R \equiv 3 \pmod 4$, and $c = \psi_{A_*}(R) > 0$. For every prescribed $N > 0$, there is a fixed positive pair (I, f) such that*

$$f^2 - \Delta I^2 = 1, \qquad 4Nc^2 \mid I,$$

and, for this same pair, every residue class $a_0 \pmod I$ is represented by an integer T for which

$$V = f^2(c - R + 4IT) - c, \qquad (\Delta I)^2 V^2 - ((\Delta I)^2 - 1)y^2 = 1 \tag{19}$$

has an integer solution y.

Proof. The Pell recurrence matrix has determinant one, so it has finite order modulo $4Nc^2$. Choose a positive even multiple m of that order and put

$$I = \psi_{A_*}(m), \qquad f = \chi_{A_*}(m).$$

Then $4Nc^2 \mid I$, I is even, f is odd, and $\gcd(f, I) = 1$.

We first justify the valuation fact

$$v_p(I) \geq v_p(m) \qquad (p \mid I). \tag{20}$$

For an odd prime dividing Δ, the recurrence reduces to $\psi_{A_*}(j) \equiv jA_*^{j-1} \pmod p$, so the first zero index is p. For an odd prime not dividing Δ, the norm-one group in the split quadratic algebra or quadratic field over $\mathbf{F}_p$ has order dividing $p - 1$ or $p + 1$, respectively; the first zero index is therefore prime to p. If $p \mid \psi_{A_*}(j)$, binomial expansion of the corresponding norm-one

unit shows that multiplying j by p raises the second-coordinate valuation by exactly one, while multiplying it by a number prime to p leaves that valuation unchanged. In the first-zero case the initial valuation is at least one, proving (20) in both odd-prime cases. At $p = 2$, because A_* is even and m is even, the doubling identity gives

$$v_2(\psi_{A_*}(m)) = v_2(m) + v_2(A_*).$$

This proves the remaining case, and hence $\gcd(m, I^2) \mid I$.

Write $\alpha = A_* + \sqrt{\Delta}$, so $\alpha^m = f + I\sqrt{\Delta}$. In the integer-coefficient quadratic algebra modulo f^2, its norm-one relation gives

$$\alpha^{4m} \equiv 1 - 4f\alpha^m \pmod{f^2}, \qquad \alpha^{4mf} \equiv 1 \pmod{f^2}.$$

The second congruence follows by raising the first to the positive power f. Thus $\lambda = 4mf$ is a period of both Pell coordinates modulo f^2. Since $\gcd(f, I) = 1$ and (20) holds,

$$\gcd(\lambda, 4I^2) \mid 4I.$$

For any specified class $a_0 \pmod{I}$, the two congruences

$$L \equiv R \pmod{\lambda}, \qquad L \equiv R - 4Ia_0 \pmod{4I^2} \tag{21}$$

are therefore compatible. Choose a positive solution L. It is $3 \pmod 4$. Put $\rho = \Delta I$ and

$$V_0 = \chi_\rho(L)/\rho, \qquad y = \psi_\rho(L).$$

The quotient V_0 is integral because the index is odd.

For completeness, the exact odd Chebyshev polynomials $Q_j(X) = \chi_b(2j+1)/b$, $X = b^2$, satisfy

$$Q_0 = 1, \quad Q_1 = 4X - 3, \quad Q_{j+1} = (4X - 2)Q_j - Q_{j-1}.$$

Induction gives

$$Q_j(X) \equiv (-1)^j(2j+1) \pmod{4X}, \qquad Q_j(1 - A_*^2) = (-1)^j\psi_{A_*}(2j+1).$$

Since $\rho^2 = \Delta(f^2 - 1) \equiv 1 - A_*^2 \pmod{f^2}$, these identities and (21) imply

$$V_0 \equiv -c \pmod{f^2}, \qquad V_0 \equiv -R + 4Ia_0 \pmod{4I^2}. \tag{22}$$

Here $L \equiv 3 \pmod 4$ makes $(L-1)/2$ odd; the period λ identifies $\psi_{A_*}(L)$ with c modulo f^2.

Because $c - R$ is divisible by four and $f^2 = 1 + \Delta I^2$, the integer

$$N_0 = V_0 + c - f^2(c - R)$$

is divisible by f^2 and congruent to $4Ia_0 \pmod{4I^2}$. The coprimality $\gcd(f^2, 4I^2) = 1$ now gives

$$T = \frac{N_0}{4If^2} \in \mathbb{Z}, \qquad T \equiv a_0 \pmod{I},$$

where the last step also uses $f^2 \equiv 1 \pmod I$. Rearrangement supplies the first equation in (19); the Pell identity for $\chi_\rho(L), \psi_\rho(L)$ supplies the second. The initial pair (I, f) depended only on N and the canonical data, not on a_0. □

Remark 5.2. The squared-modulus period is $4mf$, not $4m$. The latter is generally incorrect. Also, even I and odd f are choices made for completeness; soundness did not restrict I to be even. These two details are recorded explicitly in the inherited independent arithmetic audit.

We can now finish completeness for both new equations. Starting with a finite ZFC refutation, use the inherited canonical data (18). Lemma 4.2 selects t with $P > 0$. If $P = 1$, choose $s = 1, g = 0$. If $P \geq 2$, then $3P - 4 > 0$ is 2 (mod 3) and hence nonsquare; ordinary Pell existence gives positive s, g with

$$g^2 - (3P - 4)s^2 = 1.$$

Apply Lemma 5.1 with $N = s$, then put $r = I/(sc^2) \in \mathbb{Z}$. At this stage the pair (I, f) and the coordinate r are fixed.

By intersectivity, choose a_0 with $\pi(a_0 + \gamma) \equiv 0 \pmod{I}$. The interpolation lemma supplies $T \equiv a_0 \pmod{I}$ and its integer auxiliary root y, without changing I, f, r. The main canonical root d satisfies

$$A + Bd = \Phi_H(HT + d - b_0) = H^5\pi(T + \gamma),$$

so $(A + Bd)/S$ is an integer. Choose

$$n = \frac{A + Bd}{S} + y + 3f + 9\tau + 27g. \tag{23}$$

One conjugate factor in (8) then vanishes. For (9), replace y in (23) by δy. Every choice is made from already fixed data: native coordinates, t, s, the pair (I, f) and r, the intersective residue, (T, y), and finally n. There is no circular dependence of the modulus on the requested residue.

Together with Section 4, this proves that each new polynomial has an integer zero if and only if there is a finite syntactic ZFC refutation, with the inherited ordinary calibration identified above. The elementary arithmetic arguments and those calibration proofs are formalizable in ZFC, yielding (1). This does not supply a new machine-checked internal ZFC derivation of the complete biconditional.

6 The literal denominator-free norm

We give a finite ring recipe for the denominator clearing in (8). This also specifies exactly what the executable `norm5.py` computes. Use generic arguments $x, a_1, a_2, d_0, e_0, N_0, \delta_0$, and work temporarily in the free rank-four algebra with basis $1, u_0, v_0, u_0v_0$ and relations $u_0^2 = d_0$, $v_0^2 = e_0$. For a four-tuple $z = (z_0, z_1, z_2, z_3)$, its square has coordinates

$$\begin{aligned}
[z^2]_0 &= z_0^2 + d_0z_1^2 + e_0z_2^2 + d_0e_0z_3^2,\\
[z^2]_1 &= 2(z_0z_1 + e_0z_2z_3),\\
[z^2]_2 &= 2(z_0z_2 + d_0z_1z_3),\\
[z^2]_3 &= 2(z_0z_3 + z_1z_2).
\end{aligned}$$

Let $\mathbf{e} = (1, 0, 0, 0)$ and define

$$\begin{aligned}
\mathbf{t} &= (x^2 + d_0 + e_0, -2x, -2x, 2),\\
\mathbf{q} &= \delta_0(\mathbf{t} - (a_1 + a_2)\mathbf{e}) - N_0\mathbf{e},\\
r_0 &= \delta_0^2a_1a_2, \qquad s_0 = \delta_0N_0,\\
\mathbf{h} &= \mathbf{q}^2 - 4(r_0 + s_0(a_1 + a_2))\mathbf{e},\\
\mathbf{f} &= \mathbf{h}^2 - 64r_0s_0\mathbf{t}.
\end{aligned}$$

If $\mathbf{f} = (f_0, f_1, f_2, f_3)$, put

$$\begin{aligned}
g_0 &= f_0^2 + e_0f_2^2 - d_0(f_1^2 + e_0f_3^2),\\
g_1 &= 2(f_0f_2 - d_0f_1f_3),\\
\mathcal{C}(x; a_1, a_2, d_0, e_0, N_0, \delta_0) &= g_0^2 - e_0g_1^2.
\end{aligned}$$

Every operation in this recipe is addition, subtraction, or multiplication in the coefficient ring. No root, division, or data-dependent loop occurs in the exported arithmetic circuit.

The three-root identity

$$F_8(x;a,b,c) = \big((x^2-a-b-c)^2 - 4(ab+ac+bc)\big)^2 - 64abcx^2$$

shows that $\mathbf{f}$ represents $\delta_0^4 F_8(x-u_0-v_0;a_1,a_2,N_0/\delta_0)$. The final two quadratic norms therefore give the universal polynomial identity

$$\mathcal{C} = \delta_0^{16}\mathcal{F}_{32}(x;a_1,a_2,d_0,e_0,N_0/\delta_0). \tag{24}$$

The right side is interpreted after clearing denominators, so the identity is a polynomial identity even at $\delta_0 = 0$. For the principal construction substitute

$$(x,a_1,a_2,d_0,e_0,N_0,\delta_0) = (Z, B^2D_{\mathrm{m}}, 9S^2D_{\mathrm{f}}, 81S^2D_\tau, 729S^2D_{\mathrm{g}}, S^2C_{\mathrm{y}}, \delta).$$

For the companion, replace the last two arguments by $(S^2\delta C_{\mathrm{y}}, 1)$.

The independent exact checker constructs F_{32} by successive binomial translations $F(x+\sqrt{a})F(x-\sqrt{a})$ and compares every integer coefficient after denominator clearing. All 20349 terms match (24). Its intermediate term counts are $2, 6, 35, 495, 20349$. This is an exact universal polynomial check; the additional modular evaluations in the release serve as transcription controls for the specialized front end.

7 Attained degree and arithmetic cost

The degrees of the relevant registers are

Register	S	A	B	D_{m}	D_{f}	D_τ	D_{g}	δ	V	C_{y}
Degree	34	35	28	14	20	34	46	32	25	82

The norm $\mathcal{F}_{32}$ is even in Z and homogeneous of degree 16 in Z^2 and the five radicands. Consequently its cleared version is a homogeneous polynomial of degree 16 in

$$\delta Z^2, \quad \delta B^2 D_{\mathrm{m}}, \quad 9\delta S^2 D_{\mathrm{f}}, \quad 81\delta S^2 D_\tau, \quad 729\delta S^2 D_{\mathrm{g}}, \quad S^2 C_{\mathrm{y}}.$$

These six input degrees are, respectively,

$$102, \quad 102, \quad 120, \quad 134, \quad 146, \quad 150.$$

The coefficient of the sixteenth power of the last input is one. Therefore $(S^2C_{\mathrm{y}})^{16}$ is the entire degree-2400 contribution; every other contribution has smaller degree. This yields the degree upper bound and reduces attainment to a single leading-form check.

Write

$$\begin{aligned} q_0 &= 6\beta j, & W_{00} &= 12w,\\ a_0 &= 2q_0^4 W_{00} v = 2^7 3^5 \beta^4 j^4 wv, & I_0 &= src^2,\\ R_0 &= -q_0^3(W_{00}C - q_0 z), & b &= 4I_0 T - R_0. \end{aligned}$$

The leading forms of $\Delta, H, Q, D_{\mathrm{f}}, V$ are respectively $a_0^2, 4a_0, a_0^4 I_0^2, a_0^2 I_0^2, a_0^2 I_0^2 b$. It follows that the leading form of P_{2400} is exactly

$$2^{320} a_0^{288} I_0^{128} b^{32}. \tag{25}$$

Selecting the maximal T power in b^{32} gives the explicit coefficient

$$\big[j^{1152} w^{288} v^{288} s^{160} r^{160} c^{320} T^{32}\big] P_{2400} = 2^{2400} 3^{1440} \beta^{1152} > 0. \tag{26}$$

The exponents sum to 2400. Since no other fixed port occurs in this coefficient, specializing to the actual six-tuple cannot make it vanish.

For the companion, the radicand $S^2\delta C_{\mathrm{y}}$ has degree 182, uniquely above the other radicands and Z^2. Its sixteenth power is the degree-2912 part, with leading form

$$2^{320}a_0^{352}I_0^{160}b^{32}.$$

In particular,

$$[j^{1408}w^{352}v^{352}s^{192}r^{192}c^{384}T^{32}]P_{2912} = 2^{2848}3^{1760}\beta^{1408} > 0. \tag{27}$$

The exponents sum to 2912. These are attained degrees of the specialized polynomials, not only syntactic degree estimates for circuits with generic coefficient ports.

The final live-node census is

Polynomial	Additions	Multiplications	Subtractions	Total
P_{2400}	44	139	32	215
P_{2912}	44	133	32	209

All declared variables and all six coefficient leaves are live. The counts include the complete denominator clearing and the complete norm, not just the arithmetic front end. A multiplication by a fixed scalar is counted, as is each square; already computed registers may be reused. The exporter removes unused registers and coalesces identical operations. No assertion of optimal circuit complexity is made.

8 A direct-calculus quartic in 1190 integers

8.1 The literal endpoint and its source

The second construction improves a different part of the frontier. It starts from the received synchronized equation in 522 nonnegative coordinates [8], which directly encodes the numerical calculus C_{222}. It uses neither K215 nor Jones's universal polynomial. All coefficients of the final polynomial are ordinary small integers.

Let $R_0, \dots, R_{599}$ be the entire ordered list of quadratic residuals in `direct/output/equation.json`, on the stated coordinates $x_0, \dots, x_{1189}$. Define the one ordinary polynomial

$$Q_{\mathrm{direct}}(x_0, \dots, x_{1189}) = \sum_{j=0}^{599} R_j(x_0, \dots, x_{1189})^2, \qquad x_i \in \mathbb{Z}. \tag{28}$$

The adjacent file `expanded_polynomial.json` gives this same polynomial as a complete list of 26,411 collected monomials. A term $[c, [i_1, \dots, i_s]]$ means $c\prod_{r=1}^{s} x_{i_r}$; the empty index list is the constant monomial. Every coefficient is a literal integer. There is no remaining power with variable exponent, factorial, binomial, variable-length product, or unspecified coefficient.

Theorem 8.1 (Direct integer endpoint)**.** *The polynomial in* (28) *has exact degree four, uses all 1190 displayed coordinates, and has coefficient height* 83 900 628*. The largest absolute coefficient in any residual is* 4556*. Composing the arithmetic equivalences below with the ordinary inherited* C_{222} *calibration proves*

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{1190}\ Q_{\mathrm{direct}}(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

The frozen natural parent has 522 coordinates, 400 quadratic residuals, 20,225 collected monomials, and height 83 900 628. Its exact source file and the complete received package are included under `direct/source/` and `direct/upstream/synchronized/`. The new reduction preserves all 522 coordinates and 400 residuals until the integer-domain conversion is complete.

In particular, it modifies no proof-calculus rule, shared Pell frame, CRT relation, or coefficient of the natural parent.

The parent represents a finite C_{222} derivation through a commutative Horn flow, a finite accumulator path, CRT certificates, and synchronized Pell graphs. Its global proof constructs compatible witnesses for a common binomial radix, one power frame, and common auxiliary Pell parameters. Those simultaneous-existence theorems are inherited. Our new transformation begins after the full natural polynomial has been fixed; it does not identify independently chosen witnesses.

8.2 A finite certificate for the necessary signs

Replacing every one of the 522 natural coordinates by an individually guarded integer gives the received 2088-integer companion. Many of those guards are redundant because the equations already imply the required signs. The following elementary rule makes the redundancy explicit.

Lemma 8.2 (Scaled sign inference)**.** *Suppose a residual is*

$$cx + Q(\boldsymbol{y}) = 0, \qquad c \in \mathbb{Z} \setminus \{0\},$$

where x does not occur in Q and every coefficient of $-cQ$ is nonnegative. If all variables occurring to an odd power in a monomial of Q are nonnegative, then $x \geq 0$.

Proof. The residual implies $c^2x = -cQ$. Every even power of an integer is nonnegative. The hypotheses on odd powers and coefficients therefore make the right side nonnegative. Since $c^2 > 0$, we obtain $x \geq 0$. □

This rule proves a sign; it does not eliminate x by division or claim an integral polynomial formula for x. Its monic special case is the usual implication from $x = F(\boldsymbol{y})$ when F has nonnegative coefficients. The actual certificate has one useful nonmonic instance: a row equivalent to $16A = 20 + 4T$ proves $A \geq 0$ from $T \geq 0$. This saves one further three-coordinate guard beyond the monic-only construction.

The delivered certificate selects 234 source coordinates for explicit guards, then gives 288 applications of Lemma 8.2. Each record specifies the target, the original residual, the multiplier c^2, the entire polynomial $-cQ$, and its odd-power dependencies. The records are ordered so that every needed sign is already proved. In particular, circular implications cannot serve as unconditional nonnegativity arguments.

An independent enumeration checks that the 234 guarded coordinates are exactly those that occur as the head of no eligible rule of Lemma 8.2 in the literal residual list. Their closure reaches all 522 coordinates. The selected guard count is therefore minimal for this specific single-row inference system. That finite statement says nothing about stronger multi-row arguments, sign normalizations, or a different presentation of the polynomial.

Lemma 8.3 (Quadratic nonnegativity guard)**.** *For every $x \in \mathbb{Z}$,*

$$x \geq 0 \quad \Longleftrightarrow \quad \exists a, b, c \in \mathbb{Z} \quad x = a^2 + b^2 + c(c+1).$$

Proof. The reverse implication follows from the nonnegativity of squares and of the product of two consecutive integers. For the forward direction, $4x + 1$ is 1 or 5 modulo 8 and is not divisible by 4. Legendre's three-square theorem represents it as a sum of three integer squares. Modulo 4, exactly one representing square is odd. Reorder the terms and write

$$4x + 1 = (2a)^2 + (2b)^2 + (2c + 1)^2.$$

Subtract one and divide by four. □

For each of the 234 selected coordinates x_i, append the quadratic residual

$$x_i - a_i^2 - b_i^2 - c_i^2 - c_i = 0, \tag{29}$$

using three fresh unrestricted integer variables. The resulting checkpoint has

$$522 + 3 \cdot 234 = 1224 \quad \text{coordinates,} \qquad 400 + 234 = 634 \quad \text{quadratic residuals.}$$

Its expanded quartic has 23,590 nonzero monomials and coefficient height 83 900 628.

Proposition 8.4 (Natural/integer existence equivalence)**.** *The 1224-coordinate integer checkpoint has a zero if and only if the frozen 522-coordinate natural parent has a zero.*

Proof. At an integer zero, Lemma 8.3 makes the 234 guarded source coordinates nonnegative. Apply the 288 ordered records of the sign certificate: each is justified by Lemma 8.2 and by previously proved hypotheses. Every original coordinate is then natural. The unchanged first 400 residuals give a natural parent zero.

Conversely, start with any natural parent zero and use Lemma 8.3 independently for the 234 selected coordinates. The resulting choices satisfy all new residuals. Uniqueness of the added guard witnesses is neither claimed nor needed. □

8.3 Removing signed coordinates without losing a guard

Lemma 8.5 (Integral monic projection)**.** *Suppose an integer residual system includes $v - F(\boldsymbol{w}) = 0$ or its negative, with $F \in \mathbb{Z}[\boldsymbol{w}]$ independent of v. Substitute F for v in every other residual and delete the defining row and coordinate. Projection is a bijection on the integer zero sets, with inverse $v = F(\boldsymbol{w})$.*

Proof. Every source zero has $v = F(\boldsymbol{w})$ and therefore projects to a zero of the substituted system. Conversely, a target zero extends uniquely by the displayed value of v, which is an integer and satisfies the defining row and every original residual. □

The received exact projection algorithm [9] applies this lemma while requiring every residual to remain quadratic. Affine restorations preserve that degree bound automatically. A quadratic restoration is allowed only when its target occurs only as a scalar multiple, never in a product with another coordinate, in every other remaining row. The support cap controls expression size and never truncates a polynomial. All changed residuals are computed exactly.

The integer checkpoint admits 34 delivered substitutions: 23 affine and 11 quadratic. No unconstrained coordinate is discarded. Negative restoration coefficients are permitted over $\mathbb{Z}$. Every guard constraint is preserved either as a substituted residual or through the exact restoration formula of a deleted guard row. Its consequence is not lost merely because its original coordinate is eliminated. Apply all inverse maps in reverse order to recover the complete guarded checkpoint, then apply Proposition 8.4.

The projected equation therefore has $1224 - 34 = 1190$ coordinates and $634 - 34 = 600$ quadratic residuals. Over $\mathbb{Z}$, a sum of squares vanishes exactly when every residual vanishes. This proves existence equivalence between (28) and the frozen natural parent.

8.4 Logical calibration and the verification boundary

The parent's numerical-calculus calibration is

$$C_{222}(1) \longleftrightarrow \ \mathrm{BProv}(\mathrm{ZF}, \bot) \ \longleftrightarrow \ \mathrm{BProv}(\mathrm{ZFC}, \bot).$$

The named interfaces are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Their source copies and pins are included in the inherited

provenance. The second uses the constructible-universe equiconsistency argument. The calibration is a dependency of the parent, not a newly proved property of the 1190-variable polynomial.

Negating the composed existence equivalences proves Theorem 8.1. All new transformations are finite algebra and ordinary arithmetic formalizable in ZFC, including its internal natural and integer domains. There is no appeal to an external enumeration that works only for standard witnesses. There is also no new complete Lean/Rocq proof term or encoded object-language ZFC derivation. The result asserts the equivalence and proves neither consistency nor existence of a zero.

The independent standard-library checker imports neither the producer nor the optimizer. It authenticates the frozen parent bytes, checks the whole unchanged source prefix, all literal guards, the complete ordered sign certificate, all 34 substitutions and coordinate maps, and both expanded coefficient dictionaries. It verifies that every coordinate occurs and that degree four is attained. Four deliberately corrupted artifacts are rejected without relying on hash failure: an inference coefficient, a guard sign, a restoration, and an expanded coefficient. Normal and optimized Python runs pass. A second independent projection implementation also replays the substitutions.

The complete preserved synchronized release additionally passes its own replay, checking 88 manifest entries and regenerating 22 equation files. These finite checks establish the literal artifact identities. They do not replace the unbounded Pell/CRT theorems or the inherited calculus calibration. The full transitive proof-assistant dependency closure was not rebuilt.

8.5 Exact tradeoffs against the incoming direct equations

Construction	Integers	Rows	Terms	Height
Received synchronized lift	2088	922	26791	83901150
Received batched endpoint	1690	842	21381	2649643805758
New guarded checkpoint	1224	634	23590	83900628
New signed projection	**1190**	**600**	26411	**83900628**

Table 2: Complete direct-calculus equations over unrestricted integers. All entries have exact degree four. The two received points are taken from the recent incoming synchronized and batched archives [8, 9].

Compared with the received 1690-integer direct equation, the new endpoint saves 500 variables (29.59%) and 242 residuals (28.74%), while adding 5030 monomials (23.53%). Its coefficient height is substantially smaller. Against the synchronized integer lift of the same 522-variable natural parent, all displayed size measures improve except the unchanged degree. The 1224-coordinate checkpoint is retained as a smaller-expression tradeoff, and the natural parent itself is unchanged.

The contribution is a new explicit reduction and combination of the received constructions. The three-square theorem, sign inference, and monic projection are standard facts. We claim neither worldwide priority nor global simplicity. In particular, other repository quartics use far fewer variables with very different coefficient payloads. The improvement here is a smaller complete direct-calculus quartic with literal coefficients and an independently checkable domain conversion.

9 What the contribution changes, and what it imports

9.1 The two distinct calibration chains

For the twelve-integer equations the proof has the following form:

$$\begin{aligned}\exists \boldsymbol{x} \in \mathbb{Z}^{12}\ P_{2400}(\boldsymbol{x}) = 0 &\iff \text{the fixed source-34 native instance has a witness}\\ &\iff \mathrm{Halt}(\mathrm{K215}, \mathrm{blank})\\ &\iff \exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot).\end{aligned}$$

The first arrow is the new arithmetic reduction plus the identified signed native kernel. The later arrows are the repository's ordinary compiler and exact-table machine calibration. The companion equation has the same chain. The complete inherited directory is retained at `integer12/inherited/root_fusion/`; in particular, `provenance/CALIBRATION_STATUS.md` distinguishes the ordinary mathematical claim from the encoded proof-assistant target. The inherited proof sources include the signed native arithmetic, the finite-window tableau decoder, the source-34 coefficient adapter, and the exact K215 ordinary calibration. Their original paths, hashes, notices, and historical dependency pins are preserved.

The direct equation instead has the chain

$$\begin{aligned}\exists \boldsymbol{x} \in \mathbb{Z}^{1190}\ Q_{\mathrm{direct}}(\boldsymbol{x}) = 0 &\iff \text{the fixed 522-natural-coordinate equation has a witness}\\ &\iff C_{222}(1)\\ &\iff \exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot).\end{aligned}$$

The first arrow is established by the finite sign certificate and reversible substitutions in Section 8. The natural equation's arithmetic compiler and numerical-calculus calibration are imported with their ordinary proofs and source pins. This route makes no use of a Turing-machine transition table or a Jones universal polynomial.

The displayed implications are used as ordinary mathematical theorems formalizable in ZFC on its own natural and integer domains. Composing and negating them proves (1). No argument assumes that every ZFC theorem is true, or that a finite run has searched all proofs.

9.2 The formalization boundary is material

The supplied derivations are ordinary mathematical proofs, supplemented by finite exact certificates. No new complete Lean or Rocq development was built. In particular, an externally stated theorem about a machine and a proof predicate is distinct from a derivation of a quoted biconditional in the project's internal `SetTheory.Form` calculus. The latter requires an additional syntactic translation and formal composition for all internally coded witnesses.

The archived sources themselves record these distinctions. We preserve them instead of treating compilation, finite simulations, or hashes as a substitute for an unbounded theorem. If a reviewer declines to accept an imported ordinary calibration until independently re-establishing it, the newly checked conclusion is the exact equivalence with that calibrated parent. The final Con(ZFC) interpretation then retains the explicit calibration dependency. This is the usual dependency structure of a mathematical reduction, and it identifies the remaining work for a fully kernel-checked deliverable.

Existing formal work on Diophantine complexity bounds illustrates a different scale of machine-checked result [3]. It supplies context, not a theorem automatically validating the present literal polynomials or their coefficient instantiations.

10 A small additional improvement: twelve positive variables

The supplemental directory contains a separate optimization of the repository's ordinary twelve-positive-variable polynomial. Its variables remain positive, its exact degree remains 70, and its

zero set is unchanged. Only the evaluation schedule changes from 100 to 98 binary operations.

In the parent's unchanged registers set

$$d = c^4, \qquad e = 1 - Q, \qquad v = V + n, \qquad E_{\mathrm{a}} = QV^2 - 1, \qquad h = ev^2 + E_{\mathrm{a}}, \qquad b = eQ.$$

Then the following is an identity over the polynomial ring:

$$(dh + b)^2 - dQ(2ev)^2 = (dh - b)^2 + 4dbE_{\mathrm{a}}. \tag{30}$$

Indeed the difference between the two squared terms is $4dhb$ and $h - E_{\mathrm{a}} = ev^2$. The right side reuses E_{a} and saves two operations in the complete circuit.

This is a modest evaluator improvement. Norm polarization is already used elsewhere in the repository. It is not a new representation theorem, variable reduction, or degree result. Its complete authentic coefficient tuple is the *compact44* tuple with $\beta = 2^{590069} - 1$, preserved separately under `supplemental/ordinary98/native/`. It must not be mixed with the source-34 tuple in Section 2.

11 Assessment and concrete next research questions

The most transferable observation is the prime-valuation separator. It replaces size comparisons between radicals by congruence information that the certificate already carries. Once the least squared valuations are separated, the argument is insensitive to how large the witnesses are. It also handles an initially negative guard radicand. The use of the classical quintic is useful because its quadratic remainder has $B \equiv 1 \pmod 3$; monotonicity is no longer needed to prove $B \neq 0$.

The rational auxiliary root is a second independent improvement. Root extraction first proves it rational, the adjacent-Pell descent proves it integral, and the final equation carries only the denominator-cleared norm. This moves an integrality proof into the arithmetic argument and reduces the polynomial's degree.

The direct result suggests treating sign obligations as part of the optimization problem. A natural coordinate whose sign follows from existing equations needs no separate guard. The finite certificate makes this saving reviewable, and signed monic projection preserves it without imposing an incorrect positivity condition on eliminated coordinates.

These observations were not found in the inspected parent constructions in the forms used here. That is a repository-relative statement, not a claim of worldwide priority. The classical carrier, norm invariance, Legendre's theorem, CRT, Pell identities, and monic substitution are established mathematics. Further novelty assessment should examine earlier relation-combining and valuation-separation literature, including the broader Diophantine complexity tradition [2].

Several specific questions remain open in this packet:

1. Can a different congruence arrangement separate the same roots with a lower-degree carrier or a lower-degree scale S? Every proposed replacement must retain intersectivity, exclusion of zero-scale roots, and full residue interpolation.

2. Can the auxiliary Pell denominator be shared more efficiently with the main congruence certificate? A shorter circuit alone is not a proof that its total degree or zero set is preserved.

3. Can multi-row sign deductions or coordinate sign changes improve the direct quartic beyond the 234 guards needed by the present single-row rule family? The restricted optimality certificate leaves these stronger methods entirely open.

4. Can the arithmetic reductions be formalized against the exact internal ZFC syntax, including coefficient binding and inherited unbounded calibration? This would strengthen the verification status, even if no numerical metric improved.

No prospective bound from these questions is included in the result table.

12 Reproduction and package map

Python 3.10 or later and its standard library suffice for the arithmetic checks. A TeX distribution with the packages named in `report/report.tex` builds the article. From the package root:

```
python3 verify.py
python3 -O verify.py
cd report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The root verifier regenerates the two twelve-integer circuits and checks their exact output; checks the full generic norm identity by an independent coefficient calculation; checks the coefficient payload hashes, circuit structure and arithmetic front ends; rebuilds the direct equation and checks its full sign and projection certificates; and checks the identical-polynomial supplemental optimization. The normal and optimized runs both use explicit exceptions, not assertions that can disappear under `-O`.

The release manifest binds the deliverable files. Checks generate temporary receipts by default and do not rewrite the retained source certificates. The archived reports record the completed runs. A compact map is:

Path	Contents
`report/`	Article sources, bounded coefficient appendix, and PDF.
`integer12/`	Two new circuits, builders, independent arithmetic checks, new proof, baseline circuits, and inherited native dependencies.
`direct/`	Literal 1190-variable quartic, residuals, 1224-variable checkpoint, sign/projection certificates, independent checker, and complete synchronized parent.
`supplemental/ordinary98/`	Identical twelve-positive polynomial, 98-operation evaluator, proof, and its separate coefficient tuple.
`reviews/`	Independent assistant reviews of the new mathematics and literal direct artifacts.
`verification/`	Release receipts and PDF layout review.
`SOURCE_PINS.json`	Repository commit and received archive hashes.
`MANIFEST.json`	Final deliverable file hashes.

The primary equations are supplied as exact straight-line integer circuits because writing their fully expanded coefficient lists would obscure both the definition and the proof. A straight-line circuit with fixed integer leaves defines a unique ordinary polynomial. The direct quartic is supplied in both residual and completely expanded form.

A Complete bounded coefficient recipe

The following is the full source-34 finite arithmetic recipe [7], reproduced from the incoming root-fusion report with its licenses retained. The defining payload and source bindings are preserved under `integer12/inherited/root_fusion/coefficients/`. All ranges are bounded constants. The ordered 943 tile records $(x_{i0}, x_{i1}, x_{i2}, x_{i3})$, in west, north, east, south order, are supplied in `integer12/inherited/root_fusion/coefficients/recipe/wang943_tiles.json`. Their ordering is part of the definition: selector 0 is Start and selector 1 is Forbidden. The supplied `k215.tm` contains all 430 working actions from which the source builder reconstructs the whole ordered list. Its SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

Thus this finite table is delivered data, not an unspecified machine or a free coefficient parameter.

Put

$$b = 121, \quad L_0 = 8801, \quad d = bL_0 = 1064921, \quad \rho = 2^{121}, \quad B_0 = 2^{1064921}.$$

For $0 \le s < 4$, $0 \le j < 10$, define

$$e_s(j) = o_s + 2j, \qquad (o_0, o_1, o_2, o_3) = (1020, 1000, 980, 960),$$

and $w_j = 2^{10+2j}$ for $0 \le j < 44$. Let

$$\begin{aligned}\mathcal{E} = \{0, \dots, 942\} &\cup \{e_s(j) : 0 \le s < 4,\ 0 \le j < 10\} \\ &\cup \{1063, 1064, 1095, 1096, 2195, 3296\}.\end{aligned}$$

It has 989 elements and maximum 3296. For each tile $0 \le i < 943$, set

$$c_i = 1 + \sum_{s=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{is} + 512) w_{10s+j} + \sum_{j=40}^{43} w_j.$$

Here $\mathrm{bit}_j(m) = \lfloor m/2^j \rfloor \bmod 2$ is used only to define the finitely many fixed integer coefficients. It is not an operation on a quantified Diophantine variable. The other coefficients are

$$c_{e_s(j)} = w_{10s+j}, \quad (c_{1095}, c_{1096}, c_{2195}, c_{3296}) = (w_{40}, w_{41}, w_{42}, w_{43}), \quad c_{1063} = c_{1064} = 0.$$

Define

$$\begin{aligned}\mu &= 1022 + \sum_{j=0}^{43} w_j, \\ Z_{\mathrm{pad}} &= (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799}, \\ D_C &= \rho^{40} + \rho^{41} + \rho^{1100} + \rho^{2200} + \sum_{e \in \mathcal{E}} c_e (\rho^{4393-e} + \rho^{5495-e}), \\ D_R &= \rho, \\ M_C &= B_0 - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \ne 1}} \rho^e - 2\rho^{1063}, \\ M_{F,\mathrm{native}} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1000+2j} + \rho^{1021+2j}) + \rho^{2195} + \rho^{3296} + Z_{\mathrm{pad}}.\end{aligned}$$

The six actual fixed ports are precisely

$$\boxed{\begin{gathered}\beta = B_0 - 1, \quad K_0 = D_C + B_0 D_R, \quad M_C, \quad M_F = M_{F,\mathrm{native}} + B_0 - 1, \\ A_{\mathrm{cone}} = 2^{1113}, \qquad B_{\mathrm{cone}} = 2^{666104}.\end{gathered}}$$

All displayed exponents are fixed integers. In particular, M_F already includes its shift; shifting it again would change the equation and invalidate the binding.

The source proof verifies the whole coefficient interface, not just these sizes. It establishes local tile decoding, carry-free geometry, whole-cell rotation, mask populations, positive canonical slack, and an explicit rank controller. In particular

$$\mathrm{pc}(M_C) + \mathrm{pc}(M_{F,\mathrm{native}}) = d, \quad M_C \equiv 2 \pmod 4, \quad M_{F,\mathrm{native}} \equiv 4 \pmod 8.$$

Fixed numeral	Bit length	Minimal unsigned bytes
β	1 064 921	133 116
K_0	1 065 043	133 131
M_C	1 064 921	133 116
M_F	1 064 922	133 116
A_{cone}	1 114	140
B_{cone}	666 105	83 264
Total		615 883

Table 3: Exact fixed payload sizes before compression. The archive stores deterministic gzip encodings of the minimal unsigned little-endian bytes and binds their raw hashes.

The controller uses $d = 11^2 \cdot 13 \cdot 677$, $\lambda = 2341$, and a proved Boolean subset coverage lemma; it does not assume that d is a power of five. Its construction supplies odd canonical exponents, odd cubic scale, and the canonical parity promises in (18). The complete inherited proof and literal builder are included, with their internal source pins and licenses preserved. See `integer12/inherited/root_fusion/provenance/COEFFICIENT_PROOF.md` and `integer12/inherited/root_fusion/coefficients/recipe/source34_builder.py`. The twelve-variable transformation changes none of these six numerals.

References

[1] Andrea M. Hyde, Paul D. Lee, and Blair K. Spearman. *Polynomials* $(x^3 - n)(x^2 + 3)$ *Solvable Modulo Any Integer*. American Mathematical Monthly **121**(4) (2014), 355–358. doi:10.4169/amer.math.monthly.121.04.355. The $n = 19$ case supplies the classical intersective carrier; an elementary proof is given here.

[2] Yuri V. Matiyasevich and Julia Robinson. *Reduction of an arbitrary Diophantine equation to one in 13 unknowns*. Acta Arithmetica **27** (1975), 521–553. doi:10.4064/aa-27-1-521-553.

[3] Jonas Bayer and Marco David. *A Formal Proof of Complexity Bounds on Diophantine Equations*. In ITP 2025, LIPIcs **352**, 3:1–3:18. doi:10.4230/LIPIcs.ITP.2025.3.

[4] Vladimir Reshetnikov and ProveIt contributors. *ProveIt*, repository snapshot `85ae4070019efd66ea3e58e9a67d287528094e8b`. https://github.com/VladimirReshetnikov/ProveIt. The supplied source pins record all selected incoming archives.

[5] ProveIt incoming archive. *ConZFC Twelve Integer Interpolation*, `docs/incoming/ConZFC_Twelve_Integer_Interpolation_2026-10-24.zip`, at the snapshot in [4]. The ordinary proof `proofs/INTERPOLATION12.md` and independent audit are preserved under `integer12/proofs/`.

[6] ProveIt root-fusion package, inherited by [5]. Ordinary native/compiler proofs, normalized signed arithmetic, coefficient bindings, and calibration status. Preserved verbatim under `integer12/inherited/root_fusion/`.

[7] ProveIt source-34 coefficient construction. `integer12/inherited/root_fusion/provenance/COEFFICIENT_PROOF.md`, with authentic native input files and ordinary source copies in the same dependency tree. The finite recipe is reproduced in Appendix A.

[8] ProveIt incoming archive. *ConZFC Synchronized Certificates*, `docs/incoming/ConZFC_Synchronized_Certificates_2026-10-24.zip`, at the snapshot in [4]. Complete received package preserved under `direct/upstream/synchronized/`.

[9] ProveIt incoming archive. *ConZFC Batched Certificates*, `docs/incoming/con_zfc_batched_certificates_2026-10-23.zip`, at the snapshot in [4]. This supplies the 1690-integer comparison point and the attributed signed monic projection implementation.

[10] ProveIt ordinary-unit-products research directory, at the snapshot in [4]. Parent evaluator and ordinary proof preserved under `supplemental/ordinary98/`, with the independently pinned compact44 native inputs and inherited notices.